

Checklist for choosing a touchscreen

Considering the following factors and comparing them with the advantages and disadvantages of the key technologies, one will have a better understanding of what technology to choose:

Size. Decide the size required by your application.

Capabilities. Evaluate the type of functionality needed.

Input method. Would it be a finger, stylus or glove?

Number of points. Decide whether you want a surface with single-touch, dual-touch or multi-touch with X number of touch points.

Environment. Ascertain the conditions that the surface will be subjected to.

Durability. Know the expected life cycle of the device.

Complexity. Evaluate whether the need is for a standard product or custom designed one.

Regulatory restrictions. Are there any industry regulations that a designer needs to consider?

Availability. Ascertain availability of replacement components should a component fail.

Cost. Work out the budget for adding touch to the product design.

For more, refer to the white paper 'Choosing the Right Touch Technology,' which serves as a great reference to understanding the strengths and limitations of each technology.

—**Jamie D. Sewell, public relations & communications manager, Touch International**

market. "Hence SAW will remain in the shadow of projected-capacitive and bending-wave (or D5T) will not be able to beat optical touch," opines Sewell.

Capacitive versus optical. Being cheap, scalable and accurate, optical touchscreen technologies will be competitive.

"If you wish to do multi-touch, it requires three or four tiny sensors depending on whether you want dual-touch or three or four touch points. Capacitive-touch technology too is good, but the cost is not scalable yet up to and above 81 cm (32 inches)," says Vincent John Vincent, co-CEO, president and co-founder, GestureTek Inc. Besides, capacitive screens interfere with metal.

However, it is important to note that optical-based touchscreens suffer from occlusion and are subject to interference from sunlight that can result in false touches.

Projected infrared. "Projected infrared (IR) is in a sweet spot for cost and scalability for six and 32 touch points," feels Vincent. "This technology is relatively new but may provide an excellent solution since it scales with the size of the monitor. However, it will be a while before we see multiple inexpensive sizes above 101 cm (40 inches)," he adds.

Vincent recommends projected-capacitive-touch and optical technologies for mobile devices, projected IR and optical technology for 81cm (32

inches) and above surfaces, and optical or 3D depth cameras for offscreen touch-control surfaces.

Optical is not a good choice for mobile devices for a few reasons. First, the cameras are situated in the corners of the display, so a bezel is needed to hide these. Most mobile devices use a flush front with no bezel. (Look at the iPhone for example.) Second, in a mobile device the light falling on the screen will constantly be changing, which might cause problems like false touches or no touch detection.

Making the right pick

"With options available in the touchscreen technology space, integrators building devices with touchscreens need to choose the technology based on the use that their device or solution is going to be put to," advises Anup Tapadia, founder, TouchMagix Media.

Tapadia explains, "If the device is to be put to a more robust use, capacitive screen is the most suitable. If the surface will only be used two or three times in a day, its cost needs to be lower. For example, resistive is a cheaper option for programmable logic controllers, which use smaller touchscreens.

"However, if the user will be touching the device multiple times in a day, or the device is going to be used in harsh environments where the user touches the device or surface with

greasy fingers, capacitive or IR type of touch technology is a better option, especially if the screen size is large."

You should also know the application area of your device. Some people may want better optical clarity, some may want a robust screen, while some may want a low-cost screen. So there is no such thing as the best technology. The selection mainly depends on the use.

"Like, for large-format displays multi-touch capacitive has not evolved much, but on small format multi-touch can be seen on most touchscreen phones. So the use of a specific touchscreen technology is really where its application is," adds Taparia.

Key challenges

"One of the reasons for so many touch technologies is that there is no perfect solution," says Sewell. She enumerates a few challenges that design engineers might face while working on some of these options: "Projected-capacitive technology can be difficult to integrate, so it is important to have an experienced team available during the integration process. This technology can be sensitive to noise and electromagnetic interference (EMI) and can also emit EMI. It is important to be aware of this when designing your product. You may need to include an EMI filter, mesh or gasket into the touchscreen depending on the application."

Resistive touchscreens face the problems of low optics and a shorter lifespan. IR and optical systems have issues with occlusion and sunlight interference. It is therefore important to think about the environment in which the touchscreen will be used, to choose the right technology for the application.

While the touchscreen technologies may have matured at the hardware performance level, in terms of user experience a lot of work still needs to be done.

Tapadia observes, "In any touchscreen solution the critical aspect to work on is the combination of experience that's delivered via the touch interface. If you have a touchscreen that supports 40 touch points, an application that exploits these touch points is needed. However, what has been happening traditionally